SOLUTION

The Missionaries and Cannibals problem is solved by representing each arrangement of missionaries, cannibals, and the boat as a **state**. The solution is obtained by systematically exploring all possible valid states until the goal state is reached.

1. Represent each state as the number of missionaries, cannibals, and the boat's position on the left bank.
2. Define the initial state as **(3 Missionaries, 3 Cannibals, Boat on Left)**.
3. Define the goal state as **(0 Missionaries, 0 Cannibals, Boat on Right)**.
4. Generate all possible boat movements:
   - One missionary
   - One cannibal
   - Two missionaries
   - Two cannibals
   - One missionary and one cannibal
5. After every move, verify that:
   - The number of missionaries and cannibals remains within valid limits.
   - Missionaries are never outnumbered by cannibals on either bank unless no missionaries are present.
6. Ignore invalid or previously visited states.
7. Continue exploring valid states until the goal state is reached.
8. Display the sequence of valid crossings from the initial state to the goal state.